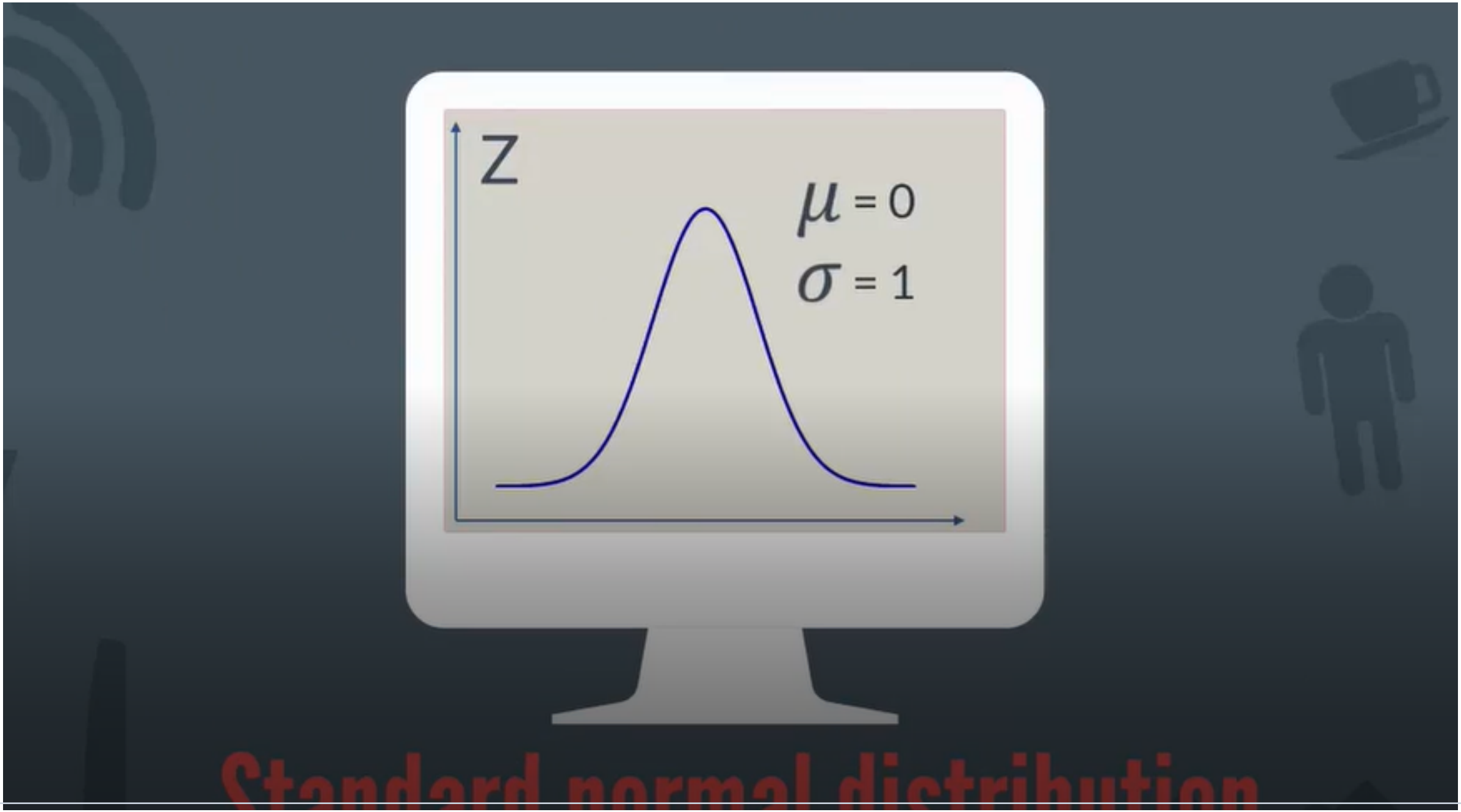

Z
$\mu = 0$
$\sigma = 1$
Standard normal distribution

From Normal to Standard normal

$N \sim (\mu\,, \sigma^2)$  $N \sim (0\,, 1\,)$

Every normal distribution can be 'standardized'

$$N \sim (\mu, \sigma^2) \longrightarrow \longrightarrow N \sim (0, 1)$$

$$\text{Standardized variable} = \frac{\text{Original variable} - \text{Mean}}{\text{Standard deviation}}$$

$$\text{z-score} = \frac{x - \mu}{\sigma}$$

$$N \sim (\mu, \sigma^2) \longrightarrow \longrightarrow N \sim (0, 1)$$



$$\text{z-score} = \frac{x - \mu}{\sigma}$$

If you take a dataset, subtract its mean from each data point and then calculate the mean once again, you will get 0.

Standard normal distribution

Standardization

Original dataset
1
2
2
3
3
3
4
4
5

Mean 3
St. dev 1.22

N~(3,1.22)

Subtract mean
-2
-1
-1
0
0
0
1
1
2

Mean 0
St. dev 1.22

N~(0,1.22)

Divide by std
x

x $\quad x - \mu \quad \frac{x - \mu}{\sigma}$

30 The standard normal distribution

Standard normal distribution

Standardization

Original dataset			Subtract mean			Divide by std		
1			-2			-1.63		
2	Mean	3	-1	Mean	0	-0.82	Mean	0.00
2	St. dev	1.22	-1	St. dev	1.22	-0.82	St. dev	1.00
3			0			0.00		
3			0			0.00		
3			0			0.00		
4	N~(3,1.22)		1	N~(0,1.22)		0.82		
4			1			0.82		
5			2			1.63		

x $\quad x - \mu$ $\quad \dfrac{x - \mu}{\sigma}$

030 The standard normal distribution

Standard normal distribution

Standardization

Original dataset
1
2
2
3
3
3
4
4
5

Mean 3
St. dev 1.22

N~(3,1.22)

Subtract mean
-2
-1
-1
0
0
0
1
1
2

Mean 0
St. dev 1.22

N~(0,1.22)

Divide by std
-1.63
-0.82
-0.82
0.00
0.00
0.00
0.82
0.82
1.63

Mean 0.00
St. dev 1.00

N~(0,1)

$$x \qquad x - \mu \qquad \frac{x - \mu}{\sigma}$$

030 The standard normal distribution